Matihus Molina Larios

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Professional Profile

I am a final-year mathematics student at National University of San Marcos with interest in Ergodic Theory, Mathematical applications to health, and Formally verified mathematics. Recently, I have developed a particular interest in the formalization of mathematics using LEAN4. I am also involved in the preparation of research papers related to machine learning applied to lung cancer and a comparative study of numerical methods for differential equations of Lane-Emden type. My current objective is to strengthen my training in pure and applied mathematics and contribute to the development of the academic community in Peru. After graduating, I aspire to pursue postgraduate studies abroad exploring different areas of mathematics.

Education

Universidad Nacional Mayor de San Marcos **Lima, Perú**
B.Sc. Student in Mathematical Sciences *03/2022 – 12/2026 (expected)*
Cumulative (CGPA): 15.19/20

Cultural Center LIDEMPERÚ **Lima, Perú**
Graduate , English Program *2018 – 2021*

Languages

Spanish: Native

English: Intermediate (B1)

Portuguese: Basic (A2)

Academic Projects

PICV-UNMSM
Proyect: "A First Encounter with Chaotic Dynamical Systems" *01/2025 – 03/2025*

- Academic advisor: Dr. Jorge L. Crisóstomo
- Study of topological conjugacy between Q_c and the Bernoulli Shift.
- Aplication of concepts from differential equation,funcional analysis,topology,and geometry.
- Click here

Course: Mathematics for ML and DS (UNMSM)
Depression Prediction in University Students *02/2025 – 03/2025*

- Application of probability ,statistics, and ML techniques .
- Supervised modeling using Python and scikit-learn.
- Click here

Stellar Structure Modeling
Numerical solution for differential equations of Lane-Emden type *10/2025 –12/2025*

- Analysis of decomposition methods and their differences.
- Numerical implementation and evaluation of the efficienty of these methods.
- Click here

Academic Events

XXXVIII Colloquium of the Peruvian Mathematical Society
PUCP – Lima *August 2025*
Attendance at talks on topology, functional analysis, differential geometry, dynamical systems, and other current research topics. This event provided an excellent opportunity to learn about the work of national and international mathematicians and to engage in discussions with distinguished researchers and professors.

E′(UBB) = 2025
UBB- Concepción *October 2025*
My first international academic experience. I greatly enjoyed the city and the opportunity to learn about a different culture. This experience motivated my interest in pursuing a master's degree abroad.

VII EICM
UNASAM- Huaráz *December 2025*
This was my first experience presenting a research project at an event of this scale. The project presented focused on stellar structure modeling via Lane–Emden type equations.

Summer School and Workshop on Optimization and Operator Learning
UNT – Trujillo *Febrary 2026*
I participated in this important event, putting my knowledge of optimization theory and programming to the test.

Research Interests

Pure Mathematics:
- Functional analysis.
- Ergodic theory.
- Mathematical foundations of artificial intelligence.
- Formally Verified Mathematics.
- Mathematical applications to human health.

Technologies:
- Programming languages: Python, R, MATLAB,LEAN4.
- Libraries: NumPy,SciPy,Pandas, Scikit-Learn, Matplotlib,PyTorch.
- Tools: Git, LaTeX.

Extracurricular Activities

Co-Director of the IEEE Computational Intelligence Society project area – UNMSM.

Former member of the cultural and artistic salsa group SALSUNI – UNMSM.

Former member of the university judo team – UNMSM

Active member of the BILDHEIT.

Certificates

Certificates of coursework,academic achievements are available in the following repository.